

Self-Buffering Antibody Formulations

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ABSTRACT: Monoclonal antibodies (mAbs) often require the development of high-concentration formulations. In such cases, and when it is desirable to formulate a mAb around pH 5.0, we explored a novel approach of controlling the formulation pH by harnessing the ability of mAbs to “self-buffer.” Buffer capacities of four representative IgG₂ molecules (designated mAb1 through mAb4) were measured in the pH 4–6 range. The buffer capacity results indicated that the mAbs possessed a significant amount of buffer capacity, which increased linearly with concentration. By 60–80 mg/mL, the mAb buffer capacities surpassed that of 10 mM acetate, which is commonly employed in formulations for buffering in the pH 4–6 range. Accelerated high temperature stability studies (50°C over 3 weeks) conducted with a representative antibody in a self-buffered formulation (50 mg/mL mAb1 in 5.25% sorbitol, pH 5.0) and with solutions formulated using conventional buffers (50 mg/mL mAb1 in 5.25% sorbitol, 25 or 50 mM acetate, glutamate or succinate, also at pH 5.0) indicated that mAb1 was most resistant to the formation of soluble aggregates in the self-buffered formulation. Increased soluble aggregate levels were observed in all the conventionally buffered (acetate, glutamate, and succinate) formulations, which further increased with increasing buffer strength. The long-term stability of the self-buffered liquid mAb1 formulation (60 mg/mL in 5% sorbitol, 0.01% polysorbate 20, pH 5.2) was comparable to the conventionally buffered (60 mg/mL in 10 mM acetate or glutamate, 5.25% sorbitol, 0.01% polysorbate 20, pH 5.2) formulations. No significant change in pH was observed after 12 months of storage at 37 and 4°C for the self-buffered formulation. The 60 mg/mL self-buffered formulation of mAb1 was also observed to be stable to freeze-thaw cycling (five cycles, –20°C → room temperature). Self-buffered formulations may be a better alternative for the development of high-concentration antibody and protein dosage forms. © 2007 Wiley-Liss, Inc. and the American Pharmacists Association *J Pharm Sci* 97:3051–3066, 2008

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INTRODUCTION

In recent years monoclonal antibody (mAb) based therapeutics have become increasingly commonplace in the repertoire of medicines available to

medical practitioners for the treatment of diseases ranging from cancer to rheumatoid arthritis.¹ However, to date, protein-based drugs have been limited to the parenteral routes of administration mainly due to their poor bioavailability via the oral route. As a result, protein drugs must be formulated as stable liquids or, in cases where liquid stability is limiting, as lyophilized dosage forms to be reconstituted with a suitable diluent prior to injection. Many critical aspects of protein formulation and parenteral manufacturing

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processes are pH sensitive; maintaining the pH of a protein drug product during manufacturing and storage is essential for its stability, efficacy, and shelf-life. For the same reasons, identifying the correct formulation pH range is of immense importance during development.

Proteins are subject to a variety of chemical degradation reactions (viz. deamidation, isomerization, hydrolysis, and oxidation) as well as physical degradation processes that include surface adsorption and irreversible aggregation. The reaction rates of these processes are often pH dependent. Moreover, both physical and chemical degradation products can lead to diminished bioactivity that affects drug potency, and have been implicated in eliciting immunological or antigenic reactions.^{2–5} The pH dependence of chemical degradation reactions in proteins has been well-documented in the literature. The rates of asparagine and glutamine deamidation are high at acidic (pH < 4) and also at basic conditions (pH > 7).^{6,7} Aspartic acid residues are known to promote the hydrolysis of adjacent peptide bonds and also undergo isomerization under acidic pH conditions.^{8,9} Disulfide scrambling reactions can be significant under neutral and basic pH conditions.¹⁰ In general, chemical degradation rates of proteins tend to be minimal in the pH range of 4.0–6.5. The physical stability of a protein, on the other hand, has a more complex, often protein-specific, dependence on pH that can be correlated to the effect of pH on the conformational and colloidal stability of the protein.^{11,12} It then stands to reason that proteins displaying adequate physical stability in the pH 4.0–6.5 range present a greater chance to be formulated as liquid dosage forms.

Buffering agents have been routinely utilized in protein drug products for pH control. In addition to providing adequate buffering capacity, an acceptable buffer for a pharmaceutical formulation must satisfy numerous requirements. The buffer must be inert and compatible with other formulation components, that is, it must not catalyze or participate in degradation reactions that may decrease the safety or efficacy of the drug. The buffering agent must be nontoxic, be well-tolerated by the patient upon administration and should not adversely affect the marketability of the drug.

Instead of employing a conventional buffer, a novel and alternative approach for controlling the pH of a high-concentration protein drug product is to use the protein to buffer the solution. Almost all

proteins have aspartic acid, glutamic acid, and histidine residues within their amino acid sequences. The side chains of these amino acids, with pK_a s of 3.7, 4.3, and 6.5 respectively,¹³ possess the ability to provide buffering action in the pH 4–6 range. We set out to determine whether, at sufficiently high concentrations, a protein's ability to buffer a solution through its solvent-exposed glutamic acid, aspartic acid, and histidine residues would be significant and comparable to the buffering capacity afforded by low-concentration conventional buffers typically used in formulations. We chose mAbs as our proteins of choice as they often necessitate the development of high-concentration liquid and lyophilized dosage forms.¹⁴

Using four IgG₂ mAbs currently under development at Amgen as examples, this report aims to demonstrate that antibodies at high concentrations may provide adequate buffering capacity for pH control in liquid dosage forms during storage and use and that self-buffered formulations may exhibit better stability profiles than their buffered counterparts. In support of this thesis, buffer capacity data as a function of antibody concentration is presented, along with accelerated and long-term stability data for self-buffered formulations of mAb1 as a representative IgG₂ molecule. The potential advantages of self-buffered formulations over conventionally buffered formulations are also discussed.

EXPERIMENTAL PROCEDURES

Materials

Bulk drug lots for mAb1—mAb4, the four mAbs under study, were received from the Amgen Process Development Group. The lots of all four mAbs were at a concentration of 70 mg/mL, and were formulated in 10 mM acetate, 5% (w/v) sorbitol (mAb1 and mAb2) or 9% sucrose (mAb3 and mAb4) at pH 5.2. All formulation excipients were of compendial (USP/EP) grade. The sorbitol was from Roquette (Keokuk, IA), the acetic and glutamic acids were from J.T. Baker (Phillipsburg, NJ) and the polysorbate 20 was from Croda (Fullerton, CA). The 1 N HCl and NaOH volumetric standards used for titration studies were purchased from the Sigma Chemical Company (St. Louis, MO).

Methods

Preparation of Acetate Standards and Antibody Solutions for Determination of Buffer Capacity

Acetate standards were prepared at 1, 2.5, 5, 7.5, 10, 15, 20, 30, and 40 mM concentrations using glacial acetic acid. The pH of each solution was adjusted to pH 5.0 using NaOH. Antibody solutions for pH titration were prepared as described below. In the case of mAb1, the bulk was first diafiltered (minimum of 7 diavolume exchanges) against a solution containing 5.25% sorbitol at pH 3.3 and then concentrated to the highest desired value (~120 mg/mL). The pH of the sorbitol solution was adjusted to pH 3.3 with 0.1 N HCl and/or NaOH. Both the diafiltration and concentration operations were conducted using the LabScale[®] tangential flow filtration, ultrafiltration/diafiltration (UF/DF) system (Millipore Corporation, Billerica, MA) fitted with three 50 cm² PELLICON[®] XL filters with BIOMAX[®] polyethersulfone membranes (MWCO—30 K) also from Millipore Corporation. Post-UF/DF, the buffer exchanged bulk was diluted to the desired concentrations (1–110 mg/mL) with the 5.25% sorbitol, pH 3.3 solution and the pH was adjusted to exactly pH 5.0 using either 0.05 N NaOH or HCl. For mAb2, the bulk was first diafiltered against a solution containing 10 mM glycine, 5% sorbitol at pH 3.8 (6 diavolumes) and then against a solution containing 5% sorbitol at pH 3.8 (7 diavolumes). As with mAb1, mAb2 solutions were prepared at various concentrations (1–60 mg/mL) using the 5.25% sorbitol, pH 3.3 solution post-UF/DF. The pH of each solution was adjusted to 5.0 using 0.05 N HCl and/or NaOH. The mAb3 and mAb4 were first dialyzed against a solution containing 5% sorbitol at pH 3.3. Dialysis was carried out at 2–8°C with a minimum sample to dialysate ratio of 1:100, with six volumes exchanges, and 5-h intervals between each exchange. Following this, concentrated stocks of the mAb3 and mAb4 bulks were prepared at 120 mg/mL using centrifugal filters. Various solutions at the desired concentrations (1–90 mg/mL) were prepared by dilution and their pH's adjusted to 5.0 using dilute (0.05 N) HCl and/or NaOH. Prior to initiating the titrations, each mAb solution was analyzed for any residual acetate using an HPLC method described later in the Methods section.

Titrations

The pH titration curves were generated over a 2-unit pH range centered ± 1 pH unit surrounding

pH 5.0. A fresh acetate or antibody solution was used for each titration. For the 1–15 mM acetate standards and mAbs, 0.2 N or 0.4 N NaOH solutions were used as titrants in the pH 5.0–6.0 range, while 0.2, 0.4 or 0.8 N HCl solutions were used for the pH 5.0–4.0 titrations. The 0.2–0.8 N HCl and NaOH solutions were prepared from the 1 N acid and base volumetric standards. Titrations of the 20–40 mM solutions were carried out using 1 N HCl (pH 4–5) or 1 N NaOH (pH 5–6) solutions. Five milliliters of the acetate buffer or antibody solution (at a given concentration) at pH 5.0 were pipetted into a 10 mL glass beaker. A stir bar was placed in the beaker and the solution was maintained under constant mixing during the entire titration period. A glass, pH electrode (Mettler Toledo—InLab[®] 423) was kept immersed in the solution and the pH was recorded using a SevenMulti[®] pH meter (Mettler Toledo, Columbus, OH). Five microliters of either the acid titrant (for the pH 5.0–4.0 range) or the base titrant (for the pH 5.0–6.0 range) were added to the solution and the new pH value was recorded. Care was taken to allow for ample equilibration time (3–4 min) and for the pH value to stabilize. All pH measurements were made in the manual mode at room temperature. Titrations were performed in triplicate for each of the 1–15 mM acetate standards and mAb1. Single titration curves were obtained for the 20, 30, and 40 mM acetate standards and for mAb2, mAb3, and mAb4.

The pH titration curves were constructed by plotting pH against the cumulative microequivalents (μEq) of acid or base added for each titration. For monoprotic and monobasic compounds, such as HCl and NaOH, the μEq measure is equal to the micromoles of titrant added. The buffer capacity (β) of a given solution was calculated as the inverse of the slope of the linear fit to the pH titration curve, normalized for the solution volume and represented as microequivalents of acid or base needed per unit change in pH per mL of the solution being titrated ($\mu\text{Eq}/\text{pH}\cdot\text{mL}$). The buffer capacity per unit acetate concentration or antibody concentration was calculated from the slopes of the linear fits to their respective buffer capacity versus concentration data and was designated by the letter “B.”

Accelerated High Temperature Stability Studies

The effect of three buffering agents (viz. acetate, glutamate, and succinate) on the stability of mAb1

was evaluated at the accelerated stability condition of 50°C over a 3-week period. All conventionally buffered formulations were at 50 mg/mL mAb1 in 5.25% sorbitol at pH 5.0. Two nominal strengths of 25 and 50 mM were chosen for each buffer type leading to a total of six formulations. The stability of the conventionally buffered formulations was compared against that of a self-buffered formulation at 50 mg/mL mAb1 and 5.25% sorbitol (to maintain isotonicity at approximately 300 ± 50 mOsm/kg), also at pH 5.0. All formulations were prepared using the dialysis method. In the case of the buffered formulations, the mAb1 bulk (60 mg/mL in 10 mM acetate 5% sorbitol, pH 5.2) was dialyzed against its respective formulation buffer (at pH 5.0) for a total of 5 exchanges. Postdialysis, the concentration of mAb1 was adjusted to 50 mg/mL and the formulation pH was adjusted to pH 5.0 using 0.05 N HCl/NaOH. The self-buffered formulation was prepared by dialyzing the mAb1 bulk against a solution containing 5.25% sorbitol at pH 3.0 (five exchanges). As with the buffered formulations, postdialysis, the mAb1 concentration was adjusted to 50 mg/mL and the formulation pH adjusted to 5.0. Each formulation was then filtered through a 0.22 μ m filter, filled into 3 mL glass vials (0.5 mL fill), stoppered with fluorotec coated stoppers (West Pharmaceutical Services, Lionville, PA) all using aseptic technique and placed on stability (50°C). The stability of mAb1 was evaluated using Size Exclusion High Performance Liquid Chromatography (SE-HPLC).

Preparation of Conventionally Buffered and Self-Buffered mAb1 Formulations for Long-Term Stability Studies

Three liquid mAb1 formulations were prepared for long-term stability studies. These included two conventionally buffered formulations with acetate and glutamate as the buffering agents along and one self-buffering formulation. All formulations were at 60 mg/mL mAb1, pH 5.2, and contained 0.01% (w/v) polysorbate 20 and were prepared using the UF/DF method. The conventionally buffered formulations contained 5% (w/v) sorbitol, while the self-buffered formulation contained 5.25% sorbitol (for isotonicity). In the case of conventionally buffered formulations, the mAb1 bulk at 70 mg/mL in 10 mM acetate, 5% sorbitol, at pH 5.2, was diafiltered for a total of 10 diavolumes against a buffer containing 10 mM glutamic acid

or 10 mM acetate and 5% sorbitol at pH 4.8, and then diluted to 60 mg/mL with the diafiltration buffer. The self-buffered formulation was prepared by diafiltration of the bulk against a solution containing 5.25% sorbitol at pH 3.3. The concentration of the diafiltered bulk was measured and diluted to 60 mg/mL (if necessary) using the respective diafiltration solutions. In the case of the conventionally buffered formulations, no further pH adjustment was necessary. The pH of the self-buffered formulation was adjusted to pH 5.2 with 0.05 N HCl and/or 0.05 N NaOH solutions. Following this, an appropriate amount of polysorbate 20 was added from a freshly prepared stock at 1% to achieve a final concentration of 0.01% in each formulation. The formulated bulk was then filtered through a 0.22 μ m filter, and filled into 1 mL capacity SCF[®], stacked-needle syringes from Becton Dickinson (Franklin Lakes, NJ). The filled syringes were shipped from Thousand Oaks, CA to Juncos, Puerto Rico and back via ground and air transportation to simulate “real-life” shipping and then placed on stability (37°C and 2–8°C).

Methods for the Determination of Acetate, Succinate, and Glutamate Concentrations

Acetate, succinate, and glutamate concentrations in various solutions were quantified using HPLC methods. For acetate and succinate quantification, a reverse phase column (Supelco LC-18, 15 cm $\times$ 4.6 mm, 3 μ m) column was employed. The mobile phase was 0.1 M phosphoric acid solution at pH 2.0, pumped at 1 mL/min for a total run time of 7 min. For glutamate quantification, a cation exchange column (Dionex CS14, 4 mm $\times$ 250 mm) was employed and the mobile phase was replaced with a 10 mM methane sulfonic acid solution at pH 2.0 also pumped at 1 mL/min for a total run time of 7 min.

Prior to loading a protein containing sample onto the HPLC system, the sample was diluted 1:1 with the mobile phase (to precipitate the protein), centrifuged using a 10 kD molecular weight cut-off centrifugal filter at 12000 RPM for 20 min, and 20 μ L of the filtrate was loaded onto the HPLC system. In all cases, chromatography was conducted using an Agilent 1100 Liquid Chromatography System with sample detection at 215 nm. A standard curve was constructed using the peak areas of known buffer (acetate, glutamate, or succinate) standards. The buffer concentration in

a given sample was determined by interpolation of the standard curve.

Size Exclusion High Performance Liquid Chromatographic Method for Evaluating the Stability of mAb1 Formulations

Size Exclusion High Performance Liquid Chromatography was used to quantitatively measure irreversible aggregation in mAb1 formulations. Chromatography was conducted using an Agilent 1100 Liquid Chromatography system and two TSK gel G3000SW_{XL} columns (TOSOH Bioscience, Montgomeryville, PA) connected in series with the column compartment temperature set to 30°C. The mobile phase contained 100 mM sodium phosphate, 500 mM sodium chloride, and 5% ethanol at pH 7.0. Thirty-five micrograms of a given sample was loaded onto the HPLC system and chromatography was carried out using a mobile phase flow rate of 0.5 mL/min for 60 min with sample detection at 235 nm.

The size exclusion chromatographic data were analyzed using either the Chromeleon[®] (Dionex Corporation, Bannockburn, IL) or ChemStation[®] (Agilent Technologies, Santa Clara, CA) software packages. The data were analyzed based on the relative areas of the main, aggregate, and clip peaks within a single chromatogram. The % sample recovery (based on the total sample peak areas relative to the $t=0$ samples) were also monitored to ensure that there was no significant protein loss over time due to adsorption or precipitation. The % sample recovery was between 90% and 110% over the course of both the long-term and accelerated studies.

RESULTS

Buffer Capacity of Acetate Standards

The ability of a buffer to resist changes in pH upon the addition of acid or base is termed its buffer capacity. A practical definition of buffer capacity (β) can be formulated as the amount of acid or base required to change the pH of a buffer solution by one pH unit.¹⁵ It can be mathematically expressed as Eq. (1) using proton balance, buffer mass balance, and dissociation constant relationships:

$$\beta_{\text{calc}} = C_T \left\{ \left[\frac{[H^+]}{K_a + [H^+]} \right]_{\text{pH}} - \left[\frac{[H^+]}{K_a + [H^+]} \right]_{\text{pH}+1} \right\} + ([H^+]_{\text{pH}} - [H^+]_{\text{pH}+1}) \quad (1)$$

In Eq. (1), C_T is the total buffer concentration and K_a is the acid dissociation constant of the buffer. The subscripts pH and pH + 1 denote any two pH values that differ by one pH unit (e.g., pH = 4, and pH + 1 = 5 for calculating β in the pH 4–5 range, and pH = 5, and pH + 1 = 6 for calculating β in the pH 5–6 range).

The pK_a of acetic acid is 4.73. Therefore, its buffering capacity is expected to be maximal at pH 4.73 and most of its buffering ability is expected to reside in a 2-unit pH range around its pK_a , from pH 3.8 to 5.8. Because the formulation pH range of interest for this study was pH 4–6, acetate was chosen as a conventional-buffer standard.

In actuality, the buffer capacity varies continuously over the pH titration curve of a buffer and only an instantaneous value at a given pH can be obtained from the tangent to the titration curve. However, over a limited pH range the

Table 1. Relevant Physicochemical Properties of the Four Antibodies under Study and the Associated Buffer Capacity Data

Antibody	pI	Number of Contributing Amino Acids/Molecule			Buffer Capacity Per mg/mL of Antibody (B^{IgG}) ($\mu\text{Eq/pH-mg}$)					
					Experimental B		B Based on Amino Acid Content		B Ratio (B/B_{calc})	
		His	Asp	Glu						
					B_{4-5} (pH 4–5)	B_{5-6} (pH 5–6)	$B_{\text{calc},4-5}$ (pH 4–5)	$B_{\text{calc},5-6}$ (pH 5–6)	$(B/B_{\text{calc}})_{4-5}$ (pH 4–5)	$(B/B_{\text{calc}})_{5-6}$ (pH 5–6)
mAb1	8.8	20	50	70	0.16	0.059	0.32	0.10	0.49	0.58
mAb2	7.5	20	52	70	0.14	0.064	0.33	0.10	0.43	0.63
mAb3	8.0	24	48	72	0.18	0.068	0.33	0.11	0.55	0.63
mAb4	8.8	20	50	66	0.14	0.052	0.31	0.10	0.46	0.53

titration curve can be approximated by a straight line, and a buffer capacity value that is constant over a range of pH values can be obtained. For this reason, two sets of pH-titration curves were generated for each acetate solution; one in the acidic direction from pH 5.0 to 4.0 and another in the basic direction from pH 5.0 to 5.5. And, the pH-titration data in the two regions could be best fit by straight lines (Fig. 1A and B, $r^2 > 0.98$). This approach enabled an important practical advantage in that the buffer capacity of acetate over the 2-unit range of pH 4–6 could be accurately expressed by two constant buffer capacity values; an acidic direction buffer capacity (β_{4-5}) for the pH 4–5 range and a basic direction buffer capacity (β_{5-6}) for the pH 5–6 range.

Another common feature of β is that it is expected to increase fairly linearly with buffer concentration. This linear relationship was observed for the acetate standards employed herein (Fig. 2, r^2 values > 0.99) with acid and basic direction B^{Acetate} (buffer capacity per unit acetate concentration) values of 0.54 and 0.39 as calculated from the slopes of the respective buffer capacity versus concentration plots. The experimentally determined B values compared well (within 10%) with the predicted values of 0.50 and 0.41 for the acid and basic directions respectively calculated using Eq. (2).

$$B_{\text{Calc}}^{\text{Acetate}} = \frac{\partial \beta_{\text{calc}}^{\text{Acetate}}}{\partial A_T} = \left[\frac{[H^+]}{K_a + [H^+]} \right]_{\text{pH}} - \left[\frac{[H^+]}{K_a + [H^+]} \right]_{\text{pH}+1} \quad (2)$$

In Eq. (2), K_a is the dissociation constant of acetate, and A_T is the total acetic acid + acetate concentration. The subscripts pH and pH + 1 denote any two pH values that differ by one pH unit (e.g., pH = 4, and pH + 1 = 5 for calculating B in the pH 4–5 range, and pH = 5, and pH + 1 = 6 for calculating B in the pH 5–6 range). Note that, because the acetate titration in basic pH range only covered a half pH unit, we calculated B^{Acetate} between 5.0 and 5.5 and normalized it to the experimentally determined value by dividing by the change in pH.

The acetate buffer capacity, determined using the above experimental procedure, was useful in two important aspects: (i) because the buffer capacities of mAbs (β^{mAb}) were determined using the same procedure and using the same titrants, their values could be quantitatively compared against those of the acetate standards, and (ii) any

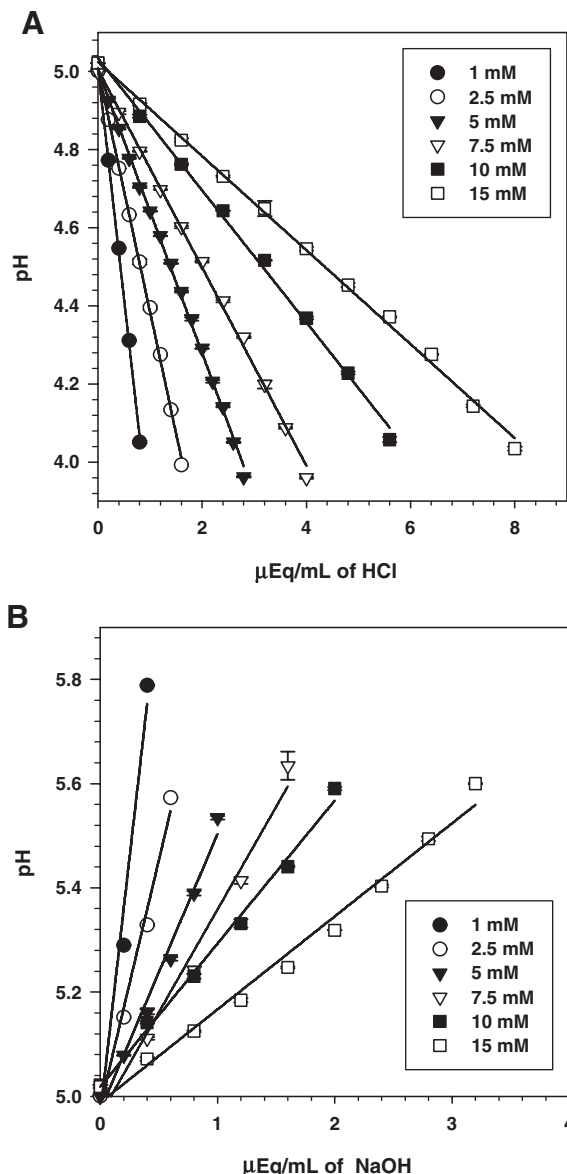


Figure 1. pH-titration of acetate solutions (1–15 mM) in the acid (panel A, pH 4.0–5.0) and base (panel B, pH 5.0–5.5) directions. Mean pH values from three independent titration measurements are presented. The error bars represent ± 1 standard deviation around the mean pH value. Titrations were conducted with 5 mL of a given acetate solution using 0.2, 0.4, or 0.8 N HCl and 0.2 or 0.4 N NaOH in the acid and basic directions respectively. All titrant volumes have been normalized to microequivalents of acid or base added per milliliter of acetate solution ($\mu\text{Eq/mL}$). The pH-titration data is sufficiently described by linear fits, represented by solid lines through each titration curve. The linear least squares coefficients of regression (r^2) values for all fits were > 0.98 .

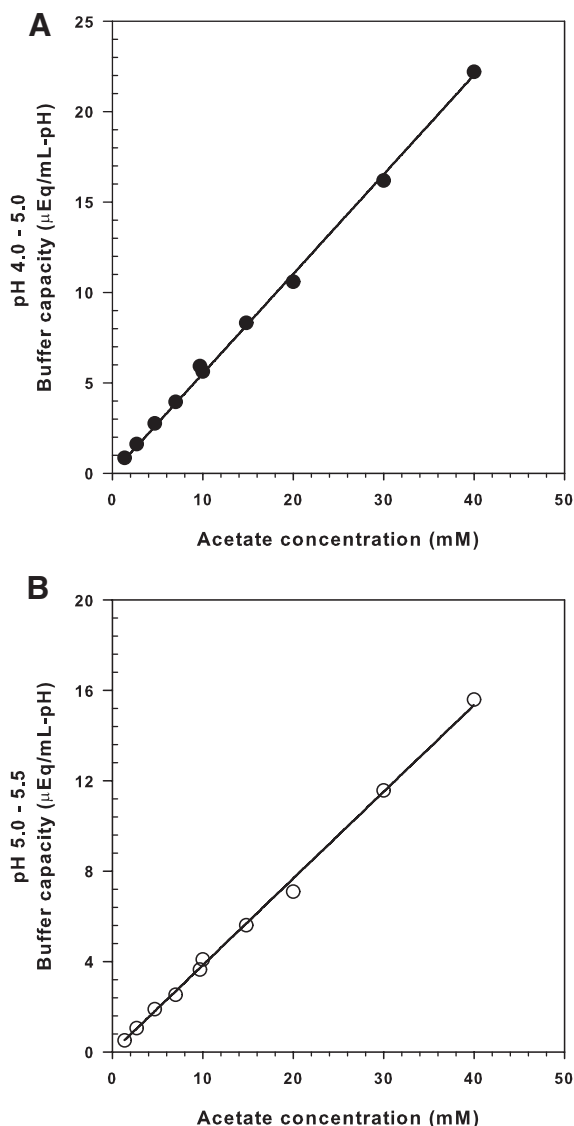


Figure 2. Buffer capacity of acetate (1–40 mM) as a function of its concentration in the pH 4.0–5.0 (panel A) and pH 5.0–5.5 range (panel B). The buffer capacity data were calculated as the inverse of the slopes of regression lines of the pH titration curves presented in Figure 1 with units of microequivalents of acid or base added per milliliter of acetate solution per pH unit ($\mu\text{Eq/mL-pH}$). The buffer capacity of acetate was observed to increase linearly with concentration. The solid lines represent linear least squares regression fits to the buffer capacity data with the coefficients of regression (r^2) values of >0.99 for both the acid and basic direction buffer capacity. From the slopes of the regression lines, buffer capacities per unit acetate concentration were calculated to be 0.54 and 0.39 $\mu\text{Eq/mL-pH-mM}$ in the acid and basic directions respectively.

buffer capacity contribution from residual acetate in a self-buffering mAb solution could be subtracted to obtain an “antibody only” buffer capacity value.

Buffer Capacity of Antibodies

Similar to acetate, the pH titration data of the mAbs in the pH 4–5 and 5–6 regions could be best fit by straight lines as shown for mAb1 in Figure 3. While some curvature is observed in the higher concentration (>60 mg/mL) titration data, the r^2 values of the linear fits were still very good (>0.95) for all data sets. The buffer capacity vs. concentration plots for mAb1 (Fig. 4) reveal that the buffer capacity of mAb1 increases linearly with concentration just as would be expected of a conventional buffer. It should be noted that the mAb1 solutions contained varying levels of residual acetate ranging from 0 (1 mg/mL) to 1.8 mM (110 mg/mL). The buffer capacity contribution of residual acetate could be removed from the measured β (β^{measured}) value of the mAb solutions using Eq. (3),

$$\beta^{\text{mAb}} = \beta^{\text{measured}} - [A_R] \times B^{\text{Acetate}} \quad (3)$$

where $[A_R]$ is the residual acetate concentration, B^{Acetate} the appropriate acidic or basic direction buffer capacity per unit acetate concentration, and β^{measured} the experimentally measured buffer capacity of a given mAb solution in the acidic or basic direction. As seen from Figure 4A and B, (open symbols), the residual acetate was observed to make a minimal contribution ($\sim 6\%$ at 110 mg/mL mAb1) to the buffer capacity of the mAb1 solutions. The subtraction procedure described by Eq. (3) was validated experimentally by measuring the acidic and basic direction buffer capacities of 60 mg/mL mAb1 solutions which contained no (<0.1 mM) residual acetate (this was achieved via exhaustive dialysis). These data appear as inverted solid triangles in Figure 4 and fit well with the residual acetate subtracted data.

Importantly, the results indicate that mAb1 possesses a significant amount of inherent buffer capacity, which at 60 mg/mL, is equivalent to approximately 15 mM acetate in the acidic direction and 10 mM acetate in the basic direction. By 110 mg/mL, the mAb1 solution acts like a 20 mM acetate buffer solution in the pH 4–6 range. Linear increases in the buffer capacity as a function of mAb concentration were also observed for mAb2, mAb3, and mAb4 (Fig. 5, panels A and C), the three other mAbs under study. While some

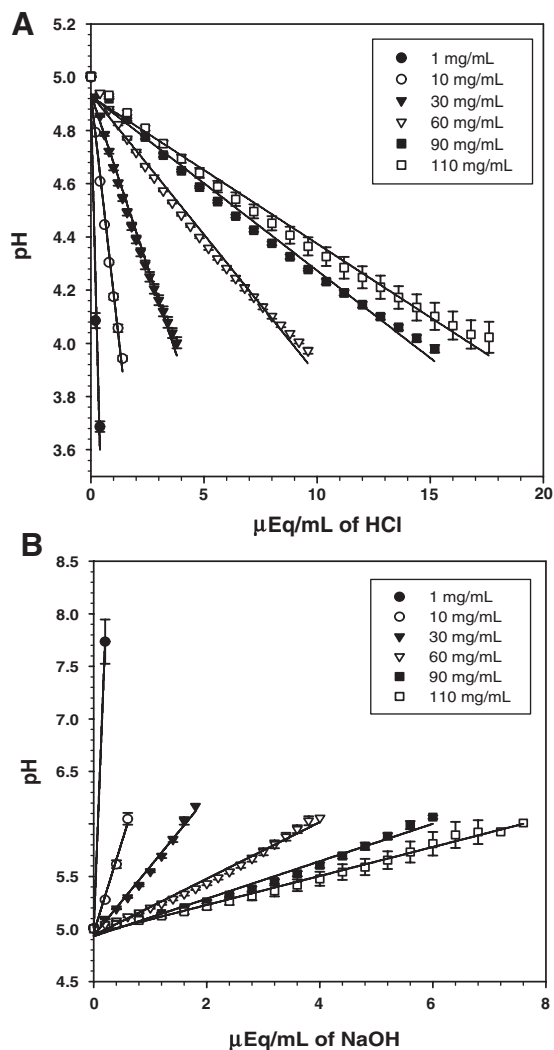


Figure 3. pH-titration of mAb1 solutions (1–110 mg/mL) in the acid (panel A, pH 4.0–5.0) and base (panel B, pH 5.0–6.0) directions. Mean pH values from three independent titration measurements are presented. The error bars represent ± 1 standard deviation around the mean pH value. Titrations were conducted using 5 mL of a given mAb1 solution with 0.2, 0.4, or 0.8 N HCl and 0.2 or 0.4 N NaOH in the acid and basic directions respectively. All titrant volumes have been normalized to microequivalents of acid or base added per milliliter of acetate solution ($\mu\text{Eq/mL}$). The pH-titration data is sufficiently described by linear fits, represented by solid lines through each titration curve. The linear least squares coefficients of regression (r^2) values for the 1 mg/mL fits were 0.95, and were >0.98 for the 10–110 mg/mL concentrations for both the acid (A) and base (B) direction data.

differences were observed in the buffer capacities of the four mAbs, the differences were relatively modest (22–25%) based on their *B* values which ranged from 0.14 to 0.18 and 0.053 to 0.068 in the

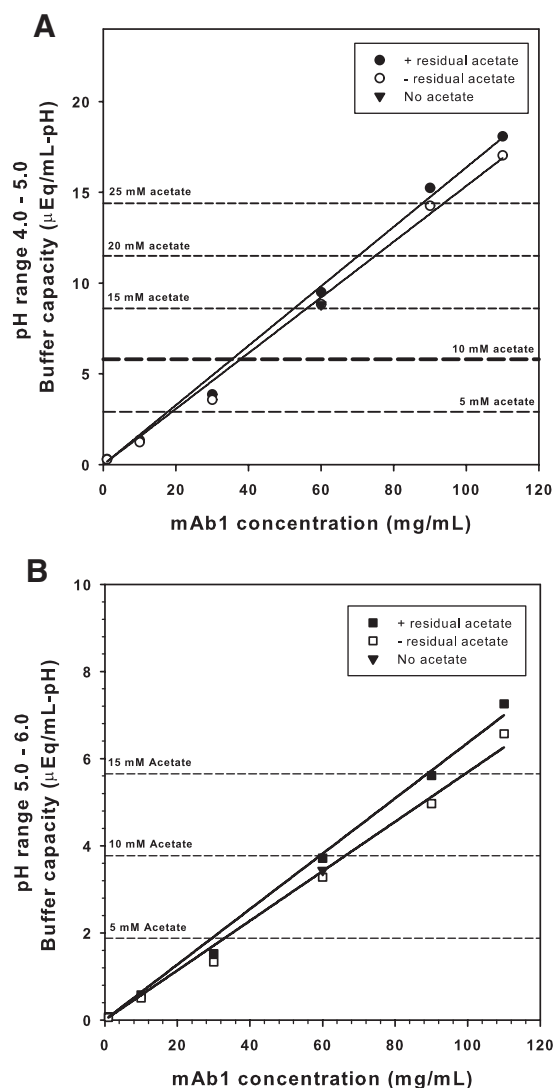


Figure 4. Buffer capacity of mAb1 as a function of its concentration. The acidic direction data (panel A) are for the pH 5.0–4.0 range while the basic direction data (panel B) are for the pH 5.0–6.0 range. The mAb1 concentration range was 1–110 mg/mL. The filled circles and square symbols (“+ residual acetate”) represent experimentally determined buffer capacity of mAb1 solutions each of which contained some amount of residual acetate. The buffer capacity was calculated as the inverse of the slopes of respective regression lines of the pH titration curves in Figure 3. The residual acetate levels in each mAb1 solution were quantified and their contribution was determined using the slopes of the buffer capacity-acetate concentration plots in Figure 2A and B. The residual acetate contribution was subtracted from experimentally determined buffer capacity values to yield the “– residual acetate” buffer capacity values and represented by open symbols. The experimentally determined buffer capacity of 60 mg/mL mAb1 solutions containing no acetate is represented by filled inverted triangles.

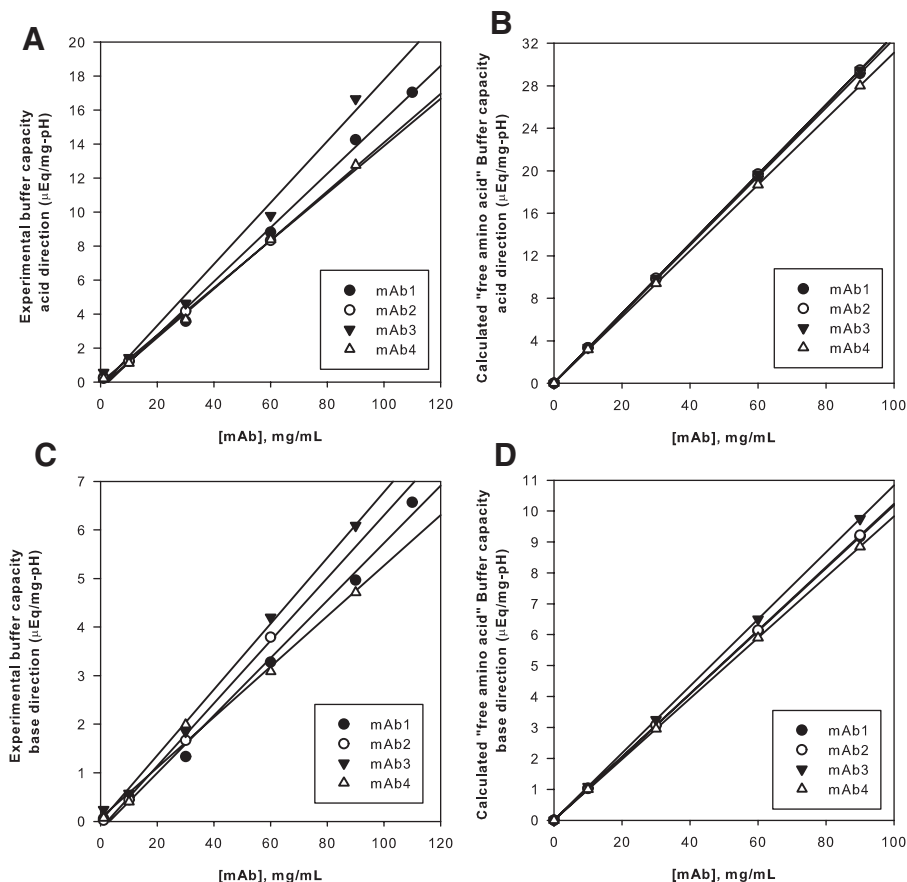


Figure 5. Comparison of the buffer capacities of antibodies. The experimental acidic direction (panel A) and the basic direction (panel C) buffer capacities of mAbs studied were obtained from the inverse of the slopes of regression lines of their respective pH titration curves (as for mAb1 in Fig. 3). The contributions of residual acetate (if any) were subtracted from these buffer capacity values. The calculated buffer capacities (panels B and D) were computed from the abundance of the contributing (Asp, Glu, and His) amino acids in each mAb and the free-state side-chain pK_a s values of the free amino acids using Equation 4. The results indicate that the experimental buffer capacities for all mAbs increase linearly with concentration ($r^2 > 0.98$) similar to that of a conventionally buffer. While the calculated buffer capacity for all the mAbs are very similar, differences are observed in the experimentally determined buffer capacity values.

acidic and basic directions respectively (Tab. 1). Of the four mAbs, mAb3 displayed the maximum buffer capacity while mAb4 had the minimum capacity. However an important point that needs to be stressed is that all four mAbs, at concentrations greater than 60–80 mg/mL, displayed buffering capacities that were comparable to 10 mM acetate, a concentration of acetate commonly employed in conventionally buffered formulations.

Accelerated High Temperature Stability Studies

Accelerated stability studies, designed to evaluate the effect of buffer type and buffer strength on the

stability of mAbs, were conducted using a representative antibody, mAb1, at 50°C over a 3-week period. The aggregation propensity of a self-buffered formulation of mAb1 at 50 mg/mL in 5.25% sorbitol, at pH 5.0 was compared to that of conventionally buffered solutions of mAb1 (50 mg/mL) formulated with 25 or 50 mM acetate, glutamate or succinate and 5.25% sorbitol at pH 5.0. The experimentally determined buffer concentrations were within 10% of their nominal values for all the buffer containing formulations while the self-buffered formulation contained 2.8 mM residual acetate. The pH of all formulations was measured at each time-point and no

significant changes were observed over the study period. The self-buffered formulation was observed to be the most resistant to aggregation when compared to any of the buffered formula-

tions over the 3-week testing period (Fig. 6A and B). Loss in % main peak area by SE-HPLC correlated to growth in the high molecular weight peak; the low molecular weight peak remained

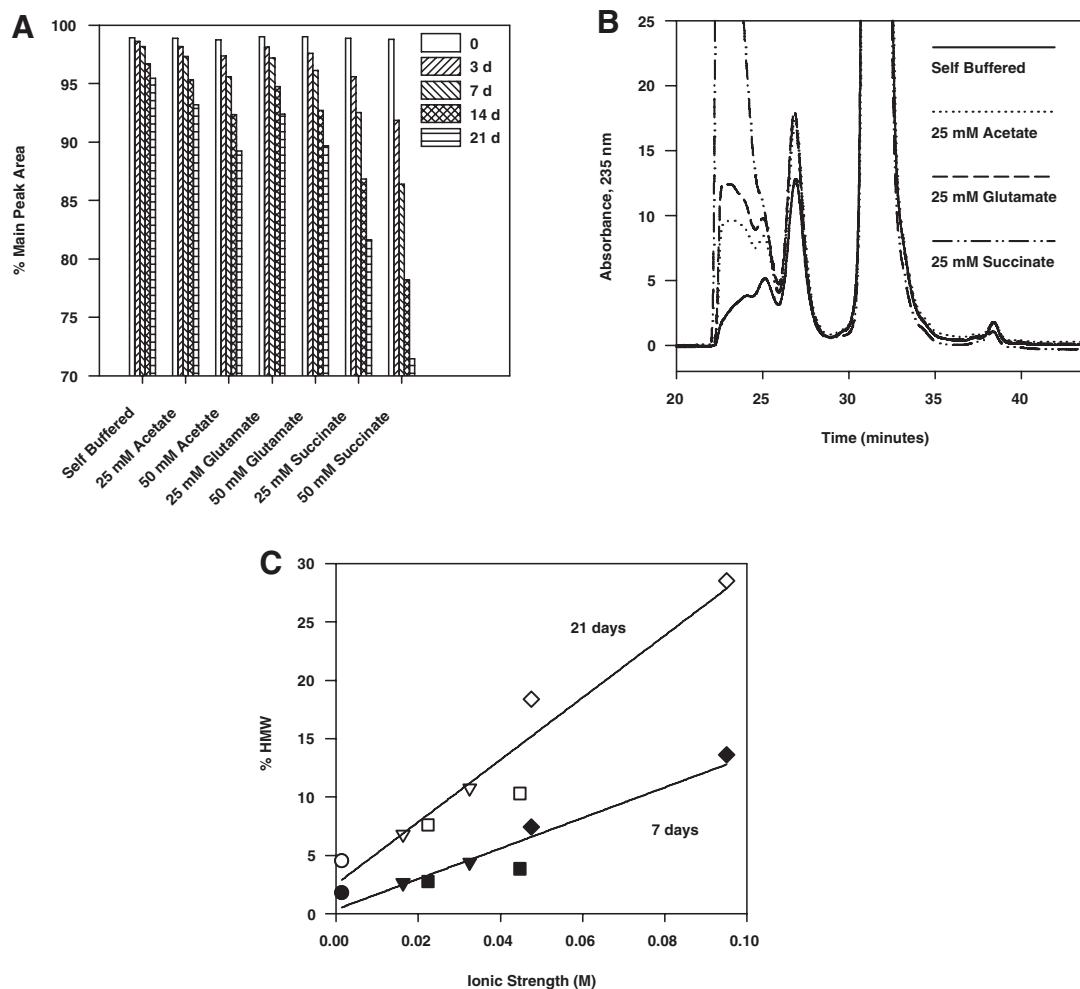


Figure 6. Comparison of the stability of acetate, glutamate, and succinate buffered and self-buffered mAb1 liquid formulations. The stability was measured in terms of aggregate formation using SE-HPLC at 50°C over a 3-week period. The data are presented as loss in percent main peak area of the three formulations over time (panel A). The buffered formulations contained 50 mg/mL mAb1 in 25 and 50 mM acetate, glutamate or succinate at pH 5.0 while the self-buffered formulation contained 50 mg/mL mAb1 in 5.25% (w/v) sorbitol at pH 5.0. The results indicate that the aggregation rate increases with buffer concentration for all the three buffers. The least amount of aggregation is observed in the self-buffered formulation. The aggregation rates in acetate and glutamate formulations are comparable and highest in the succinate formulations. An overlay of SE-HPLC chromatograms (panel B) of the 25 mM acetate, glutamate, and succinate buffered formulations, and the self-buffered formulation at the 3-week time-point indicates increased aggregation in formulations containing the conventional buffers. The formulation can be ranked in decreasing order of stability as self-buffered > acetate = glutamate > succinate. Increases in calculated ionic strength correlated well with increases in observed aggregation. Panel C includes data for the 25 and 50 mM acetate (triangle), glutamate (square), and succinate (diamond) formulations, as well as the self-buffered formulation (circle, ionic strength based on residual 2 mM acetate). R^2 values for the linear fits were 0.91 and 0.93 for the 7- and 21-day data sets, respectively.

unchanged. The aggregation rate was observed to increase with buffer concentration for all three buffers (Fig. 6A). The aggregation rates in acetate and glutamate buffered formulations were comparable but were significantly greater in succinate buffered formulations. For example, after 3 weeks of storage at 50°C, a 30% loss of main peak was observed in the 50 mM succinate formulation, an 8% loss occurred in the acetate and glutamate formulations compared to only a 4% loss in the case of the self buffered formulation. Interestingly, the mAb1 aggregation correlated well with ionic strength contributions of the buffers; the percent aggregate increased with increasing ionic strength due to the added buffer components (Fig. 6C). The ionic strength values were calculated from the concentrations of the ionized form (or forms) of the nonprotein buffer component and the sodium ion concentration resulting from bringing the acid form of the component to pH 5.0 with sodium hydroxide. The calculated strengths increased in the order self-buffered, 25 mM acetate, 25 mM glutamate, 50 mM acetate, 50 mM glutamate, 25 mM succinate, 50 mM succinate. The correlation indicates that solution ionic strength is a key factor in mAb aggregation under these accelerated conditions.

Long-Term Stability Studies

We compared the long-term stability of mAb1, at 60 mg/mL, formulated using two conventional buffers (10 mM acetate or 10 mM glutamate, 5% sorbitol and 0.01% polysorbate 20, pH 5.2) with that of a self-buffered formulation (5.25% sorbitol and 0.01% polysorbate 20, pH 5.2). The stability of the self-buffered formulation with respect to aggregation was comparable to that of the conventionally buffered formulations over a period of 6 months at 37°C and at 2–8°C (Fig. 7). The main peak area remained unchanged at 2–8°C and a ~2% loss (to dimer and higher order aggregates) was observed at 37°C over 6 months for all three formulations. No changes in pH (5.2 ± 0.05) were observed in any of the formulations (self or conventionally buffered) after 6 months of storage at 37°C or at the intended storage condition of 2–8°C (Fig. 8). The 60 mg/mL self-buffered formulation was also observed to be stable to repeated freeze-thaw cycling ($-20^{\circ}\text{C} \rightarrow$ room temperature). No change in the % main peak area (98.8%) or pH (5.20 ± 0.05) was observed after five freeze-thaw cycles.

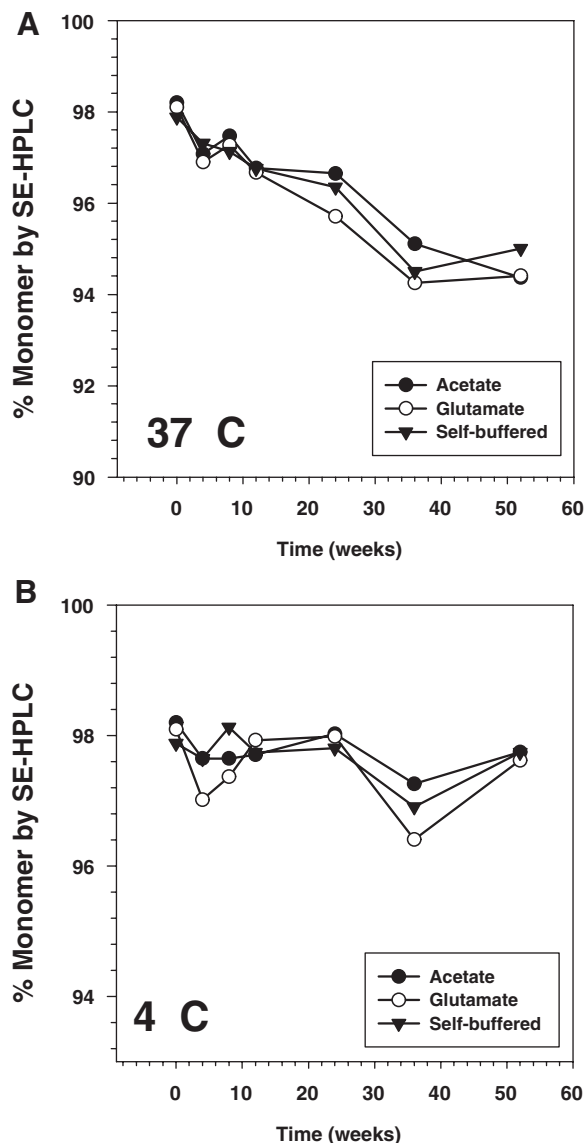


Figure 7. Comparison of the long-term stability of conventionally buffered and self-buffered 60 mg/mL mAb1 liquid formulations. The stability was measured in terms of aggregate formation using Size Exclusion-HPLC (SE-HPLC) under two temperature conditions: (i) accelerated stability condition of 37°C (panel A), and (ii) the intended storage condition of 4°C (panel B). The data are presented as loss in percent main peak area of the three formulations over time. The conventionally buffered formulations contained 10 mM acetate or 10 mM glutamate, 5% (w/v) sorbitol, and 0.01% (w/v) polysorbate 20 while the self-buffered formulation contained 5.25% (w/v) sorbitol and 0.01% (w/v) polysorbate 20 only. The results indicate a comparable loss in main peak of ~3–4% after 12 months of storage at 37°C and almost no change in main peak area at the intended storage condition of 4°C over the same time period for all three formulations.

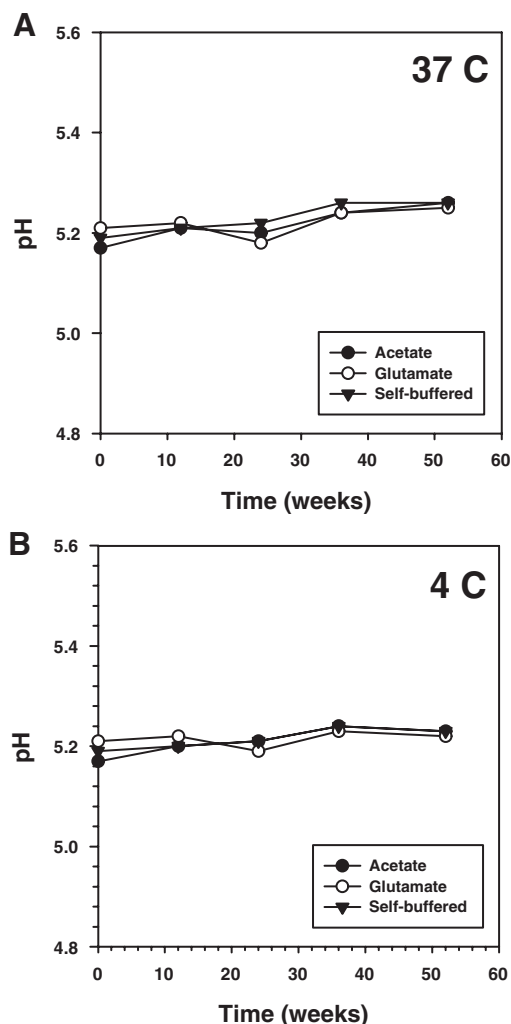


Figure 8. Comparison of pH control in conventionally buffered and self-buffered 60 mg/mL mAb1 liquid formulations. The pH was measured over a period of 12 months under two temperature conditions: (i) accelerated stability condition of 37°C (panel A), and (ii) the intended storage condition of 4°C (panel B). The conventionally buffered formulations contained either 10 mM acetate or 10 mM glutamate with 5% (w/v) sorbitol as a stabilizer/tonicity agent and 0.01% (w/v) polysorbate 20 as the surfactant while the self-buffered formulation contained 5.25% (w/v) sorbitol and 0.01% (w/v) polysorbate 20. The results indicate a stable pH for all three formulations over a 6-month period at both temperature conditions.

It should be noted that the self-buffered formulations contained a residual amount (~2 mM) of acetate. This was attributed to partitioning of acetate ions to mAb1 as a result of the combined effects of the Donnan equilibrium and the UF/DF unit operation.¹⁶ The actual buffer concentrations in the conventionally buffered formulations were

also higher than their nominal, 10 mM, values and were measured to be 17 and 13 mM for acetate and glutamate, respectively. This was attributed to the coconcentration of acetate and glutamate ions during the UF/DF process.¹⁶ An important point to note is that the buffer capacity of mAb1 in the self-buffered formulation far exceeded that of the residual acetate (Fig. 4A and B).

DISCUSSION

Buffer Capacity of Antibodies: Prediction, Measurement, and Utility

An antibody can be viewed as a polyprotic acid with its buffering action in the pH 4–6 range originating from solvent exposed aspartic acid, glutamic acid, and histidine residues. Given the high abundance of these amino acids per molecule (Tab. 1), the buffer capacity of a mAb can be expected to be significant just upon inspection of the amino acid sequence. However, prediction of the buffering capacity of an antibody (or of any protein) can be anticipated to be nontrivial. An accurate, predictive model would necessitate the knowledge of three important parameters: (i) the abundance of the three contributing amino acids, the determination of which is relatively straightforward, (ii) the number of solvent exposed Asp, Glu, and His residues, requiring a fair degree of detailed structural knowledge, and most importantly (iii) the actual pK_a s of the Asp, Glu, and His side-chains in the IgG or protein structure, which can be expected to vary significantly from their “free-state” values depending upon their local conformational environment in the molecule.¹⁷ The latter two parameters can potentially be determined using sophisticated X-ray and NMR techniques; however, such an endeavor would be extremely resource intensive. Instead, the approach and methods described in this report provide a rational, robust and relatively simple means of measuring and determining the buffer capacity levels needed for self-buffering protein formulations.

Several important inferences can be drawn from the buffer capacity data of the four mAbs presented in this report. The first is that the mAbs have a significant amount of buffering capacity. On a molar basis, the buffer capacity of just a 0.46 mM (60 mg/mL) mAb1 solution is equivalent to that of 10 mM acetate in the cumulative pH 4–6 range. Put another way, the buffer capacity of mAb1 is about 45 times that of acetate on a molar

basis. A second point is that the buffer capacity of mAbs in the acidic direction is almost twice that in the basic direction, that is, the mAbs are stronger buffers in the pH 4–5 range than in the pH 5–6 range. This is a direct result of the greater abundance of Asp and Glu residues within a given mAb compared to His (Tab. 1). A practical implication is that in the pH 4–5 range, a mAb could provide adequate buffering at even lower concentrations. For example, if formulated at pH 4.7 (with a specification limit of ± 0.3 units), the buffer capacity of a 40 mg/mL mAb1 solution would be comparable to that of 10 mM acetate.

Another point that deserves discussion is that the buffer capacities of the four mAbs were observed to be fairly comparable, within 25% of each other (Tab. 1, experimental B values). While some of the variability can be attributed to experimental error, differences are to be expected on the basis of structural differences between these molecules. In the case of IgGs of a particular isotype and origin (e.g., human IgG₂s), the major contribution to the total buffering capacity would originate from the constant regions (C_H and C_L) which can be expected to be invariant from molecule to molecule. Then, the observed differences would result from the differential abundance and/or participation (solvent exposure and pK_a changes) of contributing amino acids (Asp, Glu, and His) in the variable and hypervariable regions of the molecule. Differential abundance of amino acids appears to be an unlikely factor. The acidic and basic direction buffer capacities of the four mAbs were calculated based on their Asp, Glu and His content and their free-state, side-chain pK_a s using Eq. (4), and these were observed to be almost identical in both the acidic and basic directions (Fig. 5, panels B and D). Thus, structural differences leading to differential participation of amino acids in the buffering action appear to be more likely in explaining the observed differences.

$$\beta_{\text{calc}}^{\text{mAb}} = \sum_i N_i \times C_{\text{mAb}} \times \left\{ \left[\frac{[H^+]}{(K_a)_i + [H^+]} \right]_{\text{pH}} - \left[\frac{[H^+]}{(K_a)_i + [H^+]} \right]_{\text{pH}+1} \right\} + ([H^+]_{\text{pH}} - [H^+]_{\text{pH}+1}) \quad (4)$$

In Eq. (4), the subscript i represents the amino acids Asp, Glu, or His, N is the number of amino acid i within a given mAb, C_{mAb} is the molar mAb

concentration, and K_a is the acid dissociation constant of the amino acid side-chain. The subscripts pH and pH + 1 denote any two pH values that differ by one pH unit (e.g., pH = 4 and pH + 1 = 5 for calculating β in the pH 4–5 range).

Interestingly, the ratios of the experimental B values and the calculated B values of the mAbs (for both acidic and basic directions) averaged 0.54 with a standard deviation of 0.075 (Tab. 1). This indicates that the buffer capacity of the mAbs is only ~50% of its value based on the free amino acid content. The ratio could be brought to a value of 1 in the pH 4–5 range if aspartic acid and glutamic acid were assumed to both have a pK_a of 3.45, suggesting both side chains were acting as stronger acids than predicted. The B/B_{calc} ratio may help in providing “a rule of thumb” estimate of the buffering capacity of human IgG₂s, however this simple model needs to be tested with a larger set of antibodies.

pH Control in Formulations: How Much Buffering Capacity Is Enough?

The solution pH is a critical parameter that needs careful consideration during protein formulation development. Because both physical and chemical degradation reactions of proteins are pH dependent, protein formulations have to be designed in a narrow pH range corresponding to maximal protein stability. With respect to the pH control of a formulation, an important question arises as to how much buffering capacity is needed to adequately maintain the formulation pH during manufacturing and through its shelf-life. A quantitative assessment of the necessary buffering capacity can be extremely challenging because acid and base contamination can occur from a variety of sources, including filtration equipment, filling and transfer lines, and manufacturing vessels. While any such contamination is expected to be minimal (via rigid manufacturing and quality controls), it can lead to unacceptable pH drifts in solutions that possess very little or no buffering capacity. Another well-known source of pH drifts in liquid formulations is pharmaceutical glass containers (vials and syringes).¹⁸ Over long-term storage, sodium ions are thought to leach out from glass containers and get replaced by H^+ ions, causing the solution pH to increase.¹⁹ Formulations stored in syringes may be more prone to such pH creeps than those in vials because of the greater surface area to volume ratio of syringes

leading to an increased amount of leached sodium. It is noteworthy that the stability studies with mAb1 presented in this report were conducted with the self-buffered liquid formulation packaged in a syringe presentation and no pH drifts were observed.

If the protein itself provides adequate buffer capacity then employing a conventional buffer for pH control becomes unnecessary. The buffer capacity and long-term pH and stability data presented in this report provide solid evidence regarding the buffering ability of mAbs at high concentrations. Protein drugs have been routinely formulated using buffer concentrations of 5–20 mM.²⁰ For example, Actimmune[®] (Intermune, Brisbane, CA) is formulated using 5 mM succinate, Neupogen[®] and Neulasta[®] (Amgen, Inc., Thousand Oaks, CA) with 10 mM acetate at pH 4.0, and Avonex[®] (Biogen-Idec, San Diego, CA) with 20 mM acetate at pH 4.8. We have shown that the buffer capacity of mAbs at concentrations of 60–80 mg/mL is comparable to the levels afforded by conventional buffers in marketed protein drug products.

Challenges with Conventional Buffers: A Case for Self-Buffered Formulations

The choice of buffering agents available for use in pharmaceutical formulations in the pH range of 4–6 is limited to a handful of systems, viz. histidine, succinate, citrate, glutamate, and acetate (Tab. 2).²⁰ This is based on their pK_a s and on the available safety data for these buffers resulting from their prior use in parenteral drug products. The “precedence of use” of an excipient is a very important factor that needs consideration prior to its selection in any parenteral formulation. There are other buffers capable of providing buffering action in the pH 4–6 range, however, the lack of parenteral safety data on these compounds would likely necessitate additional toxicological and human safety studies and consequently pose significant regulatory and financial hurdles to the successful development of a drug.

The accelerated stability studies of mAb1 formulated with increasing amounts of acetate, glutamate, and succinate indicate that buffers may play an important role in mediating antibody aggregation reactions (Fig. 6). It should be noted that increased mAb1 aggregation was observed in all conventionally buffered formulations regardless of buffer type, and correlated with buffer ionic strength. Buffer identity also appears to be an

important factor as significantly lower aggregation was observed in the glutamate formulation (~9%) than in the succinate formulation (~18%) at an equivalent ionic strength of ~0.05 M. Recently, polyanionic carboxylate buffers such as succinate and citrate have been shown to form covalent adducts with the side-chain residues of proteins,²¹ and it is quite likely that such chemically modified species are also more aggregation competent. While the accelerated stability study potentially indicates an advantage of self-buffered formulations over their buffered counterparts, further studies are warranted. No stability differences were observed between conventionally and self-buffered formulations at 4 and 37°C after 12 months of storage. This may be a result of the lower concentration of conventional buffers employed in the long-term study (~15 mM) than in the accelerated study (25 or 50 mM). It is also possible that the degradation mechanisms under the accelerated conditions of 50°C are different than those at lower temperatures. Studies to elucidate the antibody degradation mechanisms in the presence of buffers are underway in our laboratory.

Conventional buffers can present additional challenges that need to be overcome during development. For example, histidine is prone to photo-oxidation in the presence of metal ions; this may lead to discoloration in formulations.²² In a study with ABX-IL8 (an IgG₂ antibody), increased aggregation and discoloration was reported in histidine-containing formulations during freeze-thaw studies of the antibody in stainless steel containers; both observations were attributed to an effect of iron ions leached from corrosion of steel containers and histidine.²³ The citrate buffer has been associated with stinging upon subcutaneous injection.²⁴ Acetate appears to be “pharmaceutically acceptable” for use in liquid protein formulations in the pH 4–6 range. Its use in lyophilized dosage forms, however, is severely limited; acetic acid is volatile and can sublime potentially causing undesirable shifts in pH during lyophilization and also upon reconstitution. Another factor to consider is the carry-over of impurities (e.g., heavy metal ions) often present in a bulk excipient and known to catalyze undesirable degradation reactions.²⁵ Self-buffered formulations, when feasible, offer the advantage of circumventing many of the issues associated with conventional buffers and provide a versatile alternative that is particularly well-suited for the development of high-concentration antibody formulations.

Table 2. Commonly Used Buffering Agents in the pH 4–6 Range and Their pK_a Values

Buffer	pK_a	Example Drug Product	Buffer Concentration (mM)
Acetate	4.73	Neupogen [®] , Neulasta [®]	10
Glutamate	4.3	Stemgen [®]	5
Succinate	$pK_{a1} = 4.18$, $pK_{a2} = 5.5$	Actimmune [®]	~5
Citrate	$pK_{a1} = 3.1$, $pK_{a2} = 4.8$, $pK_{a3} = 6.4$	Humira [®]	~6.5
Histidine	6.0	Xolair [®]	~12

LIST OF SYMBOLS AND ABBREVIATIONS

A_R	residual acetate concentration
A_T	total acetate concentration
β	buffer capacity
β_{4-5}	acidic direction buffer capacity
β_{5-6}	basic direction buffer capacity
β_{calc}	calculated buffer capacity
B	buffer capacity per unit concentration of buffering species (acetate or IgG)
B_{calc}	calculated buffer capacity per unit concentration of buffering species
C_T	total buffer concentration (M)
C_{mAb}	total monoclonal antibody (mAb) concentration (M)
IgG ₂	immunoglobulin G isotype 2
K_a	association constant (M)
mAb	monoclonal antibody
MWCO	molecular weight cut-off
N	abundance value of a given amino acid within a mAb (frequency/mAb molecule)
SE-HPLC	Size Exclusion High Performance Liquid Chromatography

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